

Forest inventory methods for estimating effects of disturbances in forests

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Project

PEPR FORESTT: National research programme
Resilience of forests
(<https://www.pepr-forestt.org/>)



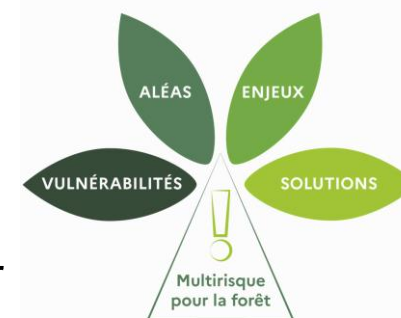
Focal project **X-RISKS**: *Analysis and management of multiple risks in forest social-ecological systems*

Work package 2: *From forest multi-risk assessment to process analysis*

Task 2.1: *Improving forest monitoring for multiple risks, their interactions and impacts on forest social-ecological systems*

Sub-task 2.1.3: *Development of sampling strategies to evaluate the impact of multiple hazards*

Post-doc project of 16 months: Alexander Massey 02/2025–



Motivation

Natural hazards cause disturbances in French / European forests

Examples of *hazards*: drought, insect or pathogen infestation, storm, wildfire

Examples of *disturbance effects*: decline of growth of trees, mortality of trees, loss of volume / biomass

Frequency and intensity of hazards increasing due to climate change

Spatial and temporal resolution of National Forest Inventories do not permit precise and timely estimation of disturbance effects

Distance between systematically placed field plots visited during one year is in order of kilometres

Need for *dedicated field measurement campaigns assisted with auxiliary information*



Drought in Germany (photo: A. Schneider)



Drought in Switzerland (photo: L. Walthert)



Bark beetles in France (photo: A. Marchi)

Auxiliary information

Auxiliary information would need to be

- (i) correlated with disturbance manifestation (presence, effect)
- (ii) timely
- (iii) available wall-to-wall (or on a dense grid of points) for sufficiently large areas

Potential *data sources*

Remote sensing (aerial photographs, airborne laser scanning, satellite images)

Environmental observations (meteorology, soil, topography)

Register databases (forest management, forest ownership)

Modelling usually needed to produce *information* from data

Vegetation height map, dominant species map, drought index map, bark beetle risk map, ...

This project: no actual auxiliary information envisaged to be used

no field measurements of perturbation effects envisaged to be carried out

Development of methodology for sampling strategy

Project idea

Assume that pertinent auxiliary information is available in map form and that potentially disturbed parts of territory can be detected

Development of methodology for *sampling strategy*: how can this auxiliary information be used for

- (i) creating a *sampling design* with which field plots are placed within territory
- (ii) improving precision of field-plot-based *estimators* of disturbance effect within territory

Disturbances of (particular) interest: spatially diffuse (with varying intensity) and discontinuous (patchy)

Example: mortality of trees due to drought

Design-based continuous-population approach

Cf. model-based approach in mapping

Design-based continuous-population approach

Forest inventory: indirect sampling of trees by placing circular plots on randomly selected points

Continuous population: points of territory of interest

Design-based approach: uncertainty in estimation arises from (hypothetical) repeated samplings of points

Sampling design (probabilistic way of selecting of points) need be taken into account in estimation

Unbiasedness: $\approx$ if sampling is repeated infinitely many times, mean of estimates obtained from samples will be equal to true value of tree population

Design-based continuous-population approach

Building blocks

Attribute density: $\approx$ sum of tree attribute values on a plot (e.g., no. of dead trees) area-prorated to attribute density value at plot centre point (e.g., no. of dead trees / hectare)

Integral of attribute density over territory is equal to sum of attribute values over trees within territory

Example: number of dead trees on a plot is converted to number of dead trees per hectare, and this value is attached to plot centre point; integral of surface of such values over territory is equal to total number of dead trees within territory

Joint probability density function of sampling points (distribution over territory)

Cordy's (1993) extension of Horvitz-Thompson theorem to sampling from continuous population

Unbiased estimator of attribute total from

- (i) attribute density values observed at sampling points
- (ii) *inclusion densities of sampling points* derived from joint probability density function

Measure of uncertainty of this estimator (unbiased estimator of variance of estimator)

Sampling design guided by auxiliary information

Deriving *joint probability density function of sampling points* *from auxiliary information map* ↪ *inclusion densities of points*

Independent identically distributed points

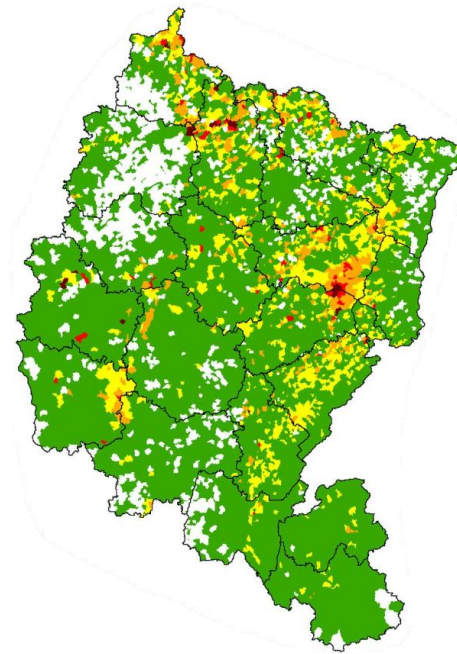
Probability density function of a point: normalised auxiliary information map

Other options: *spatially balanced sampling* from pixels of disturbed part of territory

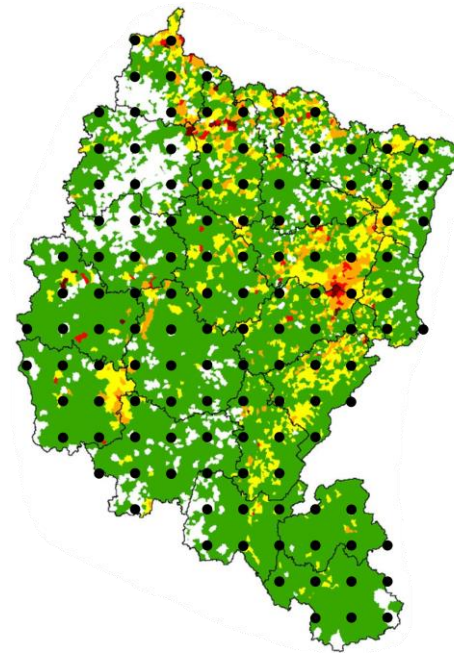
Local Pivotal Method (Grafström et al. 2012)

Generalised Random-Tessellation Stratified sampling (Stevens & Olsen 2004)

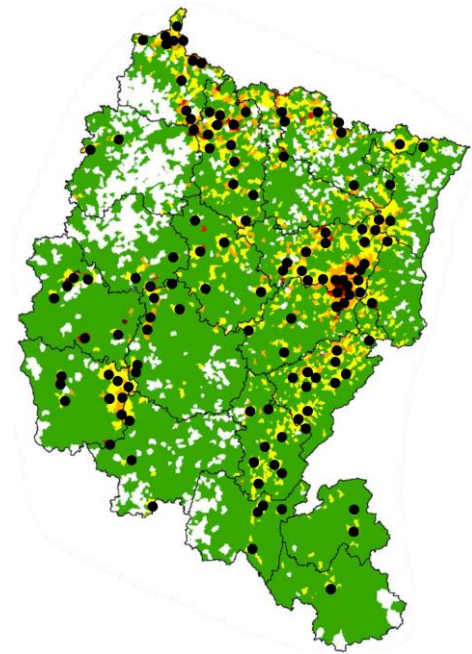
Spatially systematic stratified sampling



Auxiliary information map



Not useful: spatially systematic sampling of points over territory



Aim: sampling of points guided by auxiliary information

Model-assisted estimation with auxiliary information

Improving precision of estimation based on observations in sampling points (field plots) by using a *model between disturbance effect* (attribute density) *and auxiliary variables*

Model-assisted estimator: $\approx$ sum of model predictions in all pixels of auxiliary information map + weighted sum of model errors in field plot pixels with centre point inclusion densities as weights
Model parameters computed from same estimation data  uncertainty due to this taken into account

Cf. model-based approach in mapping, where estimator is simply sum of model predictions in pixels

Methods developed for case where *sampling points are placed uniformly randomly* within territory (Mandallaz 2013) *not applicable*

Methods permitting *non-uniform (arbitrary) placement of sampling points* developed for European-wide computation system nFIESTA

Formalisation in progress

Testing with simulations

Realistic large-scale artificial tree population created in another project: *digital twin* of a 2300-km² region in the Vosges

34.4 million trees in 16643 stands (BD Forêt polygons)

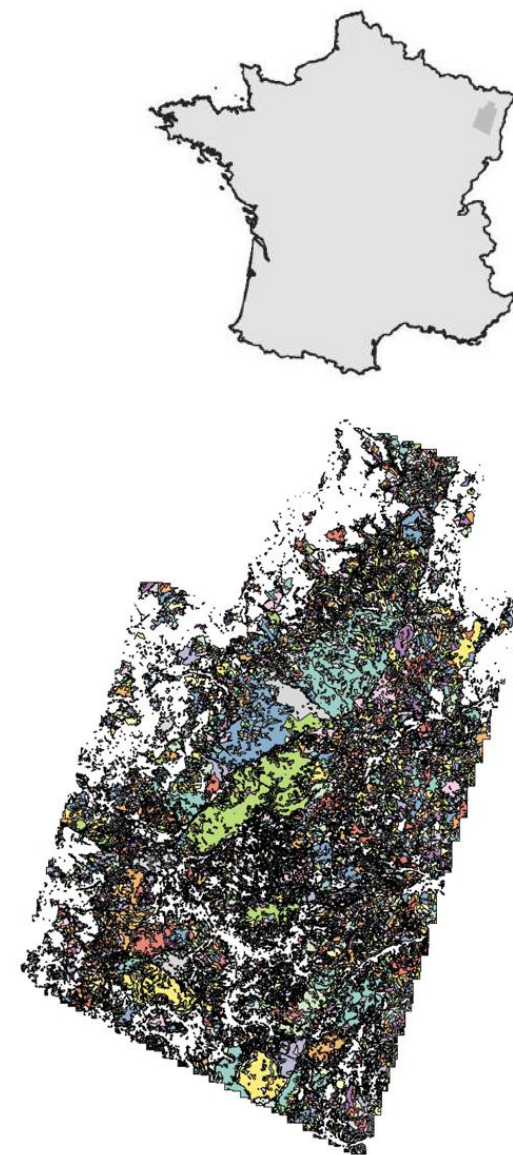
Tree attributes: location, species, stem diameter at height of 1.3 m, height, stem volume

Need be created

- (i) *tree attributes characterising disturbances* (e.g., mortality due to bark beetles or drought)
- (ii) *artificial auxiliary information sets* with different correlations to disturbance

Starting point: real bark beetle damage map (e.g., Dutrieux et al. 2020), real drought index map

Testing different sampling strategies via simulations



Figures: Sagar (2023); O. Bouriaud

Reflection on connections to French NFI

Integration to French NFI framework (sampling, field work, estimation) vs. targeted campaign separate from usual annual NFI campaign

- How to incorporate information obtained from disturbance effect estimation to standard NFI estimation?

- How to adapt annual NFI sampling design if part of field work is allocated to disturbance inventory?

Other applications

- Estimation of parameters related to rare habitats

A low-angle, upward-looking photograph of several tall birch trees. The trees have distinctive white bark with dark horizontal lenticels and thin, dark branches. Their leaves are a vibrant green, and they are set against a clear, bright blue sky. The perspective makes the trees appear to converge towards the top of the frame.

Thank you

References

- Cordy CB (1993) An extension of the Horvitz-Thompson theorem to point sampling from a continuous universe. *Statistics & Probability Letters*, 18, 353–362.
- Dutrieux R, Férét J-B & Ose K (2020). Mise au point d'une méthode reproductible pour le suivi généralisé des dégâts de scolytes par télédétection satellitaire. *ONF RenDez-Vous techniques*, 69–70, 37–44.
- Grafström A, Lundström NLP & Schelin L (2012) Spatially Balanced Sampling through the Pivotal Method. *Biometrics*, 68, 514–520.
- Mandallaz D (2013) Design-based properties of some small-area estimators in forest inventory with two-phase sampling. *Canadian Journal of Forest Research*, 43, 441–449.
- Sagar A (2023) Multisource forest inventory: A generic and flexible tool for forest resource estimation and mapping at a fine scale. PhD thesis, University of Lorraine. 178 p.
- Stevens DL & Olsen AR (2004) Spatially balanced sampling of natural resources. *Journal of the American Statistical Association*, 99, 262–278.